

Numerical Integration

A general quadrature Formula for Equidistant Ordinates:

Let $I = \int_a^b y dx$, where $y = f(x)$. Let $f(x)$ be given for certain equidistant values of x , say $x_0, x_0+h, x_0+2h, \dots$. Let the range $[a, b]$ be divided into n equal parts, each of width h so that $b-a = nh$.

Let $x_0 = a, x_1 = x_0+h = a+h, x_2 = x_0+2h = a+2h, \dots, x_n = x_0+nh = a+nh = b$. We have assumed that the $(n+1)$

ordinates $y_0, y_1, \dots, y_n$ are at equal intervals.

$$\therefore I = \int_a^b y dx = \int_{x_0}^{x_n} f(x) dx = \int_{x_0}^{x_0+nh} f(x_0+nh) dx.$$

$$= \int_{x_0}^{x_0+nh} f(x_0+nh) \cdot h du$$

since $\frac{x-x_0}{h} = u \therefore dx = h du$

$$= h \int_{x_0}^{x_0+nh} f(x_0+nh) du$$

$$= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!} \Delta^4 y_0 + \dots \right] du$$

[by Newton's difference formula] when $x = x_0$ then $u = 0$
 " $x = x_0+nh$ " $u = n$

$$= h \int_0^n \left[y_0 + u \Delta y_0 + \frac{u^2 - u}{2} \Delta^2 y_0 + \frac{u^3 - 3u^2 + 2u}{6} \Delta^3 y_0 \right. \\ \left. + \frac{u^4 - 6u^3 + 11u^2 - 6u}{24} \Delta^4 y_0 + \dots \right] du$$

$$= h \left[u y_0 + \frac{u^2}{2} \Delta y_0 + \frac{\frac{u^3}{3} - \frac{u^2}{2}}{2} \Delta^2 y_0 + \frac{\frac{u^4}{4} - u^3 + u^2}{6} \Delta^3 y_0 \right. \\ \left. + \frac{\frac{u^5}{5} - 6 \cdot \frac{u^4}{4} + \frac{11}{3} u^3 - 6u^2}{24} \Delta^4 y_0 + \dots \right]_0^n$$

$$\int_a^b f(x) dx = h \left[n y_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{6} - \frac{n^2}{4} \right) \Delta^2 y_0 \right. \\ \left. + \left(\frac{n^4}{24} - \frac{n^3}{6} + \frac{n^2}{6} \right) \Delta^3 y_0 - \left(\frac{n^5}{120} - \frac{n^4}{16} + \frac{11n^3}{72} \right. \right. \\ \left. \left. - \frac{n^2}{8} \right) \Delta^4 y_0 + \dots \right] \quad \text{--- (A)}$$

This is a general quadrature for numerical integration.

Trapezoidal Rule: Putting $n=1$ in (A), and neglecting and and higher order differences, we get

$$\int_{x_0}^{x_0+h} y dx = h \left[y_0 + \frac{1}{2} \Delta y_0 \right] = h \left[y_0 + \frac{y_1 - y_0}{2} \right] = h \left[\frac{y_0 + y_1}{2} \right]$$

$$\text{Similarly, } \int_{x_0+h}^{x_0+2h} y dx = h \left[\frac{y_1 + y_2}{2} \right], \dots$$

$$\int_{x_0+(n-1)h}^{x_0+nh} y dx = h \left[\frac{y_{n-1} + y_n}{2} \right]$$

Adding these n integrals, we get

$$\int_{x_0}^{x_0+nh} y dx = h \left[\frac{1}{2} (y_0 + y_n) + (y_1 + y_2 + \dots + y_{n-1}) \right]$$

which is known as Trapezoidal rule.

Simpson's one-third rule: Putting $n=2$ in (A) and neglecting ^{3rd} and higher order differences, we get

$$\int_{x_0}^{x_0+2h} y dx = h \left[2y_0 + 2\Delta y_0 + \frac{(8/3 - 2)}{2} \Delta^2 y_0 \right]$$

$$= h \left[2y_0 + 2(y_1 - y_0) + \frac{1}{3} (y_2 - 2y_1 + y_0) \right]$$

$$= \frac{h}{3} (y_0 + 4y_1 + y_2)$$

Similarly, $\int_{x_0+2h}^{x_0+4h} y dx = \frac{h}{3} (y_2 + 4y_3 + y_4)$

$$\int_{x_0+(n-2)h}^{x_0+nh} y dx = \frac{h}{3} (y_{n-2} + 4y_{n-1} + y_n), \text{ when } n \text{ is even}$$

Adding all these integrals, we get

$$\int_{x_0}^{x_0+nh} y dx = \frac{h}{3} \left[(y_0 + y_n) + 4(y_1 + y_3 + \dots + y_{n-1}) + 2(y_2 + y_4 + \dots + y_{n-2}) \right]$$

This is known as Simpson's $\frac{1}{3}$ rule.

Simpson's $\frac{3}{8}$ rule: Putting $n=3$ in (A), ~~and~~ and neglecting all differences above the third, we get

$$\int_{x_0}^{x_0+3h} y dx = h \left[3y_0 + \frac{9}{2} \Delta y_0 + \left(\frac{27}{3} - \frac{9}{2} \right) \frac{\Delta^2 y_0}{2!} + \left(\frac{81}{4} - 27 + 9 \right) \frac{\Delta^3 y_0}{3!} \right]$$

$$= h \left[3y_0 + \frac{9}{2} (y_1 - y_0) + \frac{9}{4} (y_2 - 2y_1 + y_0) + \frac{3}{8} (y_3 - 3y_2 + 3y_1 - y_0) \right]$$

$$= \frac{3h}{8} (y_0 + 3y_1 + 3y_2 + y_3)$$

Similarly, $\int_{x_0+3h}^{x_0+6h} y dx = \frac{3h}{8} (y_2 + 3y_4 + 3y_5 + y_6)$

$$\int_{x_0+(n-3)h}^{x_0+nh} y dx = \frac{3h}{8} (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)$$

Adding, we get

$$\int_{x_0}^{x_0+nh} y dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-1}) + 2(y_3 + y_6 + \dots + y_{n-3}) \right]$$

which is known as Simpson's $\frac{3}{8}$ rule.

Error: $-\frac{3}{8n} h^5 y^{(iv)}(\xi)$

Weddle's Rule: Putting $n=6$ in (A) and neglecting all differences of seventh and higher order, we get,

$$\int_{x_0}^{x_0+6h} y dx = h \left[6y_0 + 18\Delta y_0 + 27\Delta^2 y_0 + 24\Delta^3 y_0 + \frac{123}{10}\Delta^4 y_0 + \frac{53}{10}\Delta^5 y_0 + \frac{41}{140}\Delta^6 y_0 \right]$$

$$= \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6]$$

Similarly, $\int_{x_0+6h}^{x_0+12h} y dx = \frac{3h}{10} [y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 5y_{11} + y_{12}]$

$$\text{Error} = -\frac{h^7}{140} y^{(7)}\left(\frac{\xi}{2}\right)$$

$$\int_{x_0+(n-6)h}^{x_0+nh} y dx = \frac{3h}{10} [y_{n-6} + 5y_{n-5} + y_{n-4} + 6y_{n-3} + y_{n-2} + 5y_{n-1} + y_n] \quad (\text{if } n \text{ is a multiple of } 6)$$

Adding all these ~~terms~~ integrals, we have if n is a multiple of 6,

$$\int_{x_0}^{x_0+nh} y dx = \frac{3h}{10} [y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + y_8 + \dots]$$

This formula is known as Weddle's rule.

Ex: Calculate the value of the integral $\int_4^{5.2} \ln x dx$ by (i) Trapezoidal rule (ii) Simpson's $\frac{4}{3}$ rule (iii) Simpson's $\frac{3}{8}$ rule (iv) Weddle's rule.

After finding the true value of the integral, compare the errors in the four cases.

Solution: Taking $h=0.2$ divide the whole range of integration $[4, 5.2]$ into six equal parts. The values of $\ln x$ for each point of subdivision are given below:

x	$y = \ln x$	x	$y = \ln x$
$x_0 = 4.0$	$y_0 = 1.3862944$	$x_0 + 4h = 4.8$	$y_4 = 1.5686159$
$x_0 + h = 4.2$	$y_1 = 1.4350845$	$x_0 + 5h = 5.0$	$y_5 = 1.6094379$
$x_0 + 2h = 4.4$	$y_2 = 1.4816045$	$x_0 + 6h = 5.2$	$y_6 = 1.6486586$
$x_0 + 3h = 4.6$	$y_3 = 1.5260563$		

(i) By Trapezoidal rule, we have

$$\int_4^{5.2} \ln x dx = \frac{h}{2} [y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

$$= \frac{0.2}{2} [3.034953 + 2 \times 7.6207991]$$

$$= 1.8276551$$

(i) By Simpson's $\frac{1}{3}$ rule, we have

$$\int_4^{5.2} \ln x dx = \frac{h}{3} [y_0 + y_6 + 4(y_1 + y_2 + y_3) + 2(y_4 + y_5)]$$
$$= \frac{0.2}{3} [3.034953 + 4 \times 4.5705787 + 2 \times 3.0502204]$$

$$= 1.8278472$$

(ii) By Simpson's $\frac{3}{8}$ rule, we have

$$\int_4^{5.2} \ln x dx = \frac{3h}{8} [y_0 + y_6 + 3(y_1 + y_2 + y_4 + y_5) + 2y_3]$$
$$= 1.827847$$

(iv) By Weddle's rule, we have

$$\int_4^{5.2} \ln x dx = \frac{3h}{10} [y_0 + y_6 + 5(y_1 + y_5) + y_2 + y_4 + 6y_3]$$
$$= 1.8278474.$$

Actual value of $\int_4^{5.2} \ln x dx = [x(\ln x - 1)]_4^{5.2}$

$$= 1.8278475$$

Hence the errors due to different formulae are (i) 0.0001924 (ii) 0.00000003 (iii) 0.00000005 (iv) 0.00000001.

We observe that Weddle's rule is more accurate.

H.W: Book Ref: S.S. Sastry: Example 5.7, 5.8, 5.9

Weddle's Rule:
$$\int_{x_0}^{x_0+nh} f(x) dx = \frac{3h}{10} \left[y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + 2y_6 + 5y_7 + y_8 + 6y_9 + y_{10} + 2y_{12} + \dots + 2y_{n-6} + 5y_{n-5} + y_{n-4} + 6y_{n-3} + y_{n-2} + 5y_{n-1} + y_n \right]$$

Ex: Evaluate the value of the integral

$\int_0^1 (\sin x - \log x + e^x) dx$; by (a) Trapezoidal rule

0.2

(b) Simpson's $\frac{1}{3}$ rule (c) Simpson's $\frac{3}{8}$ rule.
After finding the actual value of the integral, compare the errors in the four cases

Ex: $\int_0^1 \frac{dx}{x^2 + 1}$ by Simpson's $\frac{1}{3}$ and $\frac{3}{8}$ rule & Weddle's rule

Ex: $\int_0^1 \frac{dx}{1+x}$

Differences: If $y_0, y_1, y_2, \dots, y_n$ denote a set of values of any function $y = f(x)$, then $y_1 - y_0, y_2 - y_1, y_3 - y_2, \dots, y_n - y_{n-1}$ are called the first differences of the function y . Denoting these differences by $\Delta y_0, \Delta y_1, \Delta y_2$ etc, we have

$$\Delta y_0 = y_1 - y_0, \Delta y_1 = y_2 - y_1, \Delta y_2 = y_3 - y_2, \dots, \Delta y_{n-1} = y_n - y_{n-1},$$

$$\Delta y_n = y_{n+1} - y_n.$$

The differences of these first differences are called second differences. Denoting them by $\Delta^2 y_0, \Delta^2 y_1$, etc,

$$\Delta^2 y_0 = \Delta y_1 - \Delta y_0 = (y_2 - y_1) - (y_1 - y_0) = y_2 - 2y_1 + y_0$$

$$\Delta^2 y_1 = \Delta y_2 - \Delta y_1 = (y_3 - y_2) - (y_2 - y_1) = y_3 - 2y_2 + y_1$$